

[standalone build] Skew-brace decomposition obstruction chapter

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Chapter 1

The decomposition obstruction of a skew brace: a bilinear class, independent of the internal residue

A skew brace is the algebraic shadow of a set-theoretic solution of the Yang–Baxter equation, and the recurring engineering question about it is the same one that runs through this whole dossier: *when does a composite come apart into its pieces?* Concretely, given a skew brace G and an ideal $D \trianglelefteq G$, when is G an internal bicrossed (Zappa–Szép) product $(G/D) \bowtie D$? This chapter answers three questions about the obstruction to that decomposition. It *names* the obstruction (a class in Rathee–Yadav skew-brace cohomology, [PEER-REVIEWED]); it *measures* the obstruction (a genuinely *bilinear* class, strictly finer than the two underlying group Schreier classes, [PROVED]); and it *isolates* the obstruction from the one datum most easily confused with it — the internal relabelling residue $[\nu]$ — by proving they are independent coordinates ([PROVED], with one surjectivity residual [COMPUTED]).

The result answers, in group-theoretic language, Ferri’s Remark 3.20 on when a quiver skew brace splits off an ideal [1], and it supplies the skew-brace / Yang–Baxter instance of the dossier’s decomposition programme. A word of warning before we begin, because the two are so easily conflated: the *existence* axis — “does G admit a Zappa–Szép factorisation at all” — is the subject of Chapter ?? and is governed by the group-cohomological class $[\omega]$. The *decomposition* axis of this chapter — “does G factor *through a prescribed ideal* D ” — is strictly finer, and its obstruction $[\Omega]$ lives in a two-operation cohomology that no single group class can see. Keeping the two apart is the point.

1.1 The three theorems

Throughout, $(G, +, \circ)$ is a skew left brace: $(G, +)$ and $(G, \circ)$ are groups sharing the identity 0, related by $a \circ (b + c) = (a \circ b) - a + (a \circ c)$, and we write $\lambda_a(b) := -a + (a \circ b)$, so $a \circ b = a + \lambda_a(b)$ and $\lambda: (G, \circ) \rightarrow \text{Aut}(G, +)$ is a homomorphism. An *ideal* $D \trianglelefteq G$ is a normal subgroup for both operations with $\lambda_a(D) = D$; it yields a short exact sequence of skew braces

$$0 \longrightarrow D \longrightarrow G \xrightarrow{\pi} H := G/D \longrightarrow 0.$$

We assume $(D, +)$ and $(D, \circ)$ are both abelian — the setting in which the extension is classified by a second cohomology — and, where a citation to Rathee–Yadav’s theory is invoked directly, the

standing hypothesis (SH) that D is a *trivial* brace ($a \circ b = a + b$ on D). A *decomposition of G through D* is an isomorphism $G \cong (G/D) \bowtie D$ onto an internal bicrossed product; equivalently (see §1.3) a sub-skew-brace complement to D .

Fix a normalised section $s: H \rightarrow G$ (so $s(0) = 0$). It determines the *good triplet*

$$\nu_h := \lambda_{s(h)}|_D, \quad \mu_h(x) := -s(h) + x + s(h), \quad \sigma_h(x) := s(h)^{-1} \circ x \circ s(h),$$

of maps $D \rightarrow D$, together with the additive and circle factor sets

$$\beta(h_1, h_2) := -s(h_1 + h_2) + s(h_1) + s(h_2) \in D, \quad \bar{\tau}(h_1, h_2) := s(h_1 \circ h_2)^{-1} \circ s(h_1) \circ s(h_2) \in D.$$

The three theorems of the chapter, stated informally now and precisely in the body:

Theorem 1.1 (Identification, [PEER-REVIEWED]; §1.3). *The obstruction to decomposing G through D is a single class $[\Omega] \in H_{\text{sb}}^2(H, D)$ in Rathee–Yadav skew-brace cohomology, and $[\Omega] = 0$ if and only if $G \cong (G/D) \bowtie D$, i.e. D has a sub-skew-brace complement. No coprimality hypothesis is needed.*

Theorem 1.2 (Bilinearity, [PROVED]; §1.4). *$[\Omega]$ is a genuinely bilinear class: the two forgetful maps to the additive and circle group Schreier classes, $\varphi_+([\Omega]) = [\beta]$ and $\varphi_\circ([\Omega]) = [\bar{\tau}]$, can both vanish while $[\Omega] \neq 0$. Vanishing of both single-operation obstructions does not imply decomposition. The obstruction is base-localised: it is decided on a single star of the ambient groupoid, and closure there propagates to all stars.*

Theorem 1.3 (Independence, [PROVED]; §§1.5–1.6). *The internal relabelling residue $[\nu] \in \prod_h \text{Aut}(D, +)/L$ (where $L := \text{im}(\lambda|_D)$) is an orthogonal direct factor of the section-independent data, independent of $[\Omega]$. Fixing $(H\text{-brace}, \mu, \sigma, L)$, the equivalence classes of extensions with a given realised residue form a torsor under a residue-independent obstruction group Ω^0 ; consequently $([\nu], [\Omega]): \text{Ext} \xrightarrow{\sim} R \times \Omega^0$ is a bijection — every realised residue co-occurs with every value of the obstruction. That every coupling-admissible residue is realised (surjectivity of the residue map) is [COMPUTED], not [PROVED]; it does not affect the product structure.*

The method is uniform and worth stating in one breath. Because $a \circ b = a + \lambda_a(b)$, once the additive group $(G, +)$ and the homomorphism λ are fixed, the circle operation — hence the factor set $\bar{\tau}$ — is *determined*, never solved for. The residue $[\nu]$ is the $D \rightarrow D$ block of λ ; the obstruction $[\Omega]$ is the off-diagonal, section-varying part. The block-triangular structure of the brace axioms then does all the work: the residue is cut out by an action-only equation that never mentions the obstruction cochain, and the obstruction is cut out by an affine equation whose homogeneous part is residue-free. Orthogonality is not a coincidence of small examples; it is the shape of the equations.

Outline. §1.2 records the factor-set calculus and the coupling equation. §1.3 proves Theorem 1.1 (identification). §1.4 proves Theorem 1.2 (bilinearity and base-localisation). §1.5 isolates the residue $[\nu]$ as an orthogonal factor and shows non-trivial kernels do not disturb $[\Omega]$. §1.6 proves the product Theorem 1.3. §1.7 gives the genuine $|G| = 16$ computational witness and states the single open residual precisely. §1.8 is the honest ledger and the relation to Ferri’s Remark 3.20.

1.2 Setup: factor sets and the coupling equation

With $(G, +)$ fixed, a skew-brace structure making D an ideal is *exactly* a homomorphism $\lambda: G \rightarrow \text{Aut}(G, +)$ satisfying $\lambda_{a \circ b} = \lambda_a \lambda_b$ (with $a \circ b := a + \lambda_a(b)$) and leaving D invariant. The circle factor set is then read off from $\circ$; it is never independently posited. This is the reduction that governs everything below: the question “does an extension with a prescribed residue exist?” is a question about λ , and the coordinates $\beta, \bar{\tau}$ are the shadow of the single identity $\lambda_{a \circ b} = \lambda_a \lambda_b$.

Definition 1.4 (Inner group and residue). The *inner group* is $L := \{\lambda_e|_D : e \in D\} = \text{im}(\lambda|_D) \leq \text{Aut}(D, +)$, with $L \cong (D, \circ)/\ker(\lambda|_D)$; write $D_0 := \ker(\lambda|_D)$, so D is a trivial brace iff $L = \{\text{id}\}$ iff $D_0 = D$. The *residue* of the section datum $\nu_h = \lambda_{s(h)}|_D$ is the coset $[\nu_h] := \nu_h L \in \text{Aut}(D, +)/L$.

In coordinates the brace axioms are equivalent to the following system (the generalised Rathee–Yadav conditions; the additive-difference circle factor set is written $T(h, k) := s(h) \circ s(k) - s(h \circ_H k) \in D$):

$$\begin{aligned}
\text{(a)} \quad & \mu_{h_1} \beta(h_2, h_3) + \beta(h_1, h_2 + h_3) = \beta(h_1, h_2) + \beta(h_1 + h_2, h_3), & \text{(a)} \\
\text{(b)} \quad & T \text{ is a circle 2-cocycle (the } \circ\text{-associativity condition),} & \text{(b)} \\
\text{(c)} \quad & \mu_h \nu_h \beta(k, l) + T(h, k +_H l) = T(h, k) - \mu_{h \circ k} \mu_h^{-1} \beta(h, -h) + \beta(h \circ k, -h) \\
& + \mu_{(h \circ k) - h} T(h, l) + \beta((h \circ k) - h, h \circ l), & \text{(c)} \\
\text{(coupling-1)} \quad & \nu_h \mu_k \nu_h^{-1} = \mu_{\lambda_h^H(k)}. & (*)
\end{aligned}$$

Here $\lambda^H : (H, \circ_H) \rightarrow \text{Aut}(H, +)$ is the induced H -brace map. Two structural facts about this system carry the whole chapter, and we flag them now:

- In equations (a)–(c), the residue datum ν appears *exactly once*, in the term $\mu_h \nu_h \beta(k, l)$ of (c).
- Equation (coupling-1) constrains the residue using only ν, μ, λ^H — it mentions neither T nor β .

This is the block-triangularity: the residue side and the obstruction side of the system decouple.

1.3 The obstruction is a skew-brace cohomology class

We first pin $[\Omega]$ down as a cohomology class and record what its vanishing means. All of this section is [PEER-REVIEWED] (referee: Rick); the cohomology machinery is Rathee–Yadav’s [2], invoked, not re-derived.

Under (SH) the pair $(\beta, \bar{\tau})$, subject to (a), (b), (c), represents a class

$$[\Omega] := [(\beta, \bar{\tau})] \in H_{\text{Sb}}^2(H, D) := Z_{\text{Sb}}^2(H, D)/B_{\text{Sb}}^2(H, D),$$

where Z_{Sb}^2 is the set of pairs satisfying the coupling (c) together with $\beta \in Z_{\text{Gp}}^2((H, +); \mu)$ and $\bar{\tau} \in Z_{\text{Gp}}^2((H, \circ); \sigma)$, and B_{Sb}^2 is generated by the section changes below. This is exactly the group that classifies skew-brace extensions inducing the fixed triplet (ν, μ, σ) . The circle datum $\bar{\tau}$ and the additive-difference factor set T of §1.2 are one datum in two coordinate systems: $\bar{\tau}$ is the circle-multiplicative form and $T = \nu_{h \circ k}(\bar{\tau})$ its additive image. We use T in the cross-equation (c) — where it must interact additively with β — and $\bar{\tau}$ for the circle cohomology class.

Lemma 1.5 (Additive coboundary; $\beta = d\theta$). *Under a section change $s'(h) = s(h) + \theta(h)$ with $\theta(0) = 0$, the additive factor set shifts by an additive coboundary and nothing else:*

$$(\beta' - \beta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2) = (d\theta)(h_1, h_2),$$

independent of $\bar{\tau}$, ν and σ .

This is the $\beta = d\theta$ convention of [2]; we adopt it verbatim. The lemma says the additive datum β is exactly the group 2-cocycle of $0 \rightarrow (D, +) \rightarrow (G, +) \rightarrow (H, +) \rightarrow 0$ with μ its conjugation action; so the forgetful assignment $\varphi_+ : [\Omega] \mapsto [\beta] \in H_{\text{Gp}}^2((H, +), D; \mu)$ is well defined — the coupling (c) constrains cocycles but not coboundaries.

Theorem 1.6 (Well-definedness of φ_+ , [PEER-REVIEWED]). *φ_+ is a well-defined homomorphism; the coupling (c) does not obstruct it. The circle forgetful map $\varphi_\circ: [\Omega] \mapsto [\bar{\tau}] \in H_{\text{Gp}}^2((H, \circ), D; \sigma)$ is defined symmetrically.*

Theorem 1.7 (Identification, [PEER-REVIEWED]). *Under (SH), for the extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ with good triplet (ν, μ, σ) ,*

$[\Omega] = 0 \iff \text{the sequence splits by a skew-brace section} \iff G \cong (G/D) \bowtie D \iff D \text{ has a sub-skew-brace complement}$

No coprimality hypothesis is required (coprimality enters only Rathee–Yadav’s stronger uniqueness theorem). The first equivalence is intrinsic to the Rathee–Yadav classification; the remaining ones are the registry result corrected-equivalence-1-2-3, via Ferri’s Theorem 3.18 [1].

A *sub-skew-brace complement* is a subset $K \subseteq G$ closed under both $+$ and $\circ$, with $K \cap D = \{0\}$ and $|K| |D| = |G|$; the equivalence records that a one-sided (additive-only or circle-only) complement is strictly weaker, which is precisely the content of the next section. The Lean corpus supplies machine-checked `decide` certificates for the descent of φ_+ and for the non-injectivity witness of §1.4 [LEAN-VERIFIED].

1.4 Bilinearity: strictly finer than both Schreier classes

The heart of the chapter’s “measurement” claim is that $[\Omega]$ is not the pair of group classes $(\varphi_+([\Omega]), \varphi_\circ([\Omega]))$; it is genuinely bilinear, seeing a coupling between the two operations that neither operation sees alone. Everything here is [PROVED].

Theorem 1.8 (The product map is not injective, [PROVED]). *Let $\Phi := (\varphi_\circ, \varphi_+): H_{\text{Sb}}^2(H, D) \rightarrow H_{\text{Gp}}^2((H, \circ), D; \sigma) \times H_{\text{Gp}}^2((H, +), D; \mu)$. Then $\ker \Phi \neq 0$: there is a skew brace with $\Phi([\Omega]) = (0, 0)$ yet $[\Omega] \neq 0$.*

The W_4 witness. Take additive group $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/4$, multiplicative group $(G, \circ) \cong \mathbb{Z}/4 \times \mathbb{Z}/2$, and ideal $D = \{0\} \times \mathbb{Z}/4$ (a trivial-brace ideal), with $H = G/D \cong \mathbb{Z}/2$. Then:

- the additive extension splits — the additive complement $\{(0, 0), (1, 0)\}$ gives $[\beta] = \varphi_+([\Omega]) = 0$;
- the multiplicative extension splits — the multiplicative complement $\{(0, 0), (1, 1)\}$ gives $[\bar{\tau}] = \varphi_\circ([\Omega]) = 0$;
- but these two complements are *distinct subsets*, and no single subset is a complement for both operations: there is no sub-skew-brace complement, so by Theorem 1.7, $[\Omega] \neq 0$.

Hence $\Phi([\Omega]) = (0, 0)$ with $[\Omega] \neq 0$. The obstruction lives in the coupling term of (c); it is *irreducibly bilinear*. $\square$

The witness is exact and small enough to enumerate; a sweep over $\mathbb{Z}/4$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/6$, $\mathbb{Z}/8$, $\mathbb{Z}/9$, $\mathbb{Z}/2 \times \mathbb{Z}/4$ produces ten W_4 -type witnesses on $\mathbb{Z}/2 \times \mathbb{Z}/4$, and locates a “phantom” family in which 96 of 144 candidate ideals admit an ambient additive complement while 48 do not — the additive picture alone cannot predict decomposition. Formally, the two forgetful functors send $[\Omega]$ to the additive and multiplicative Schreier classes $[\omega_+] \in H^2((G/D)_+, D_+)$ and $[\omega_\circ] \in H^2((G/D)_\circ, D_\circ)$, and $[\Omega]$ may be nonzero while *either or both* vanish. This is why the decomposition axis of this chapter is strictly finer than the existence axis of Chapter ??: the group class $[\omega]$ is one leg of Φ , and Φ has a kernel.

Base-localisation. In the groupoid (quiver skew brace) generalisation of the problem — where G is a groupoid with a per-star additive structure and D a wide normal sub-bundle — the obstruction is not merely finer, it is *localised*. The naive “heap-transport” conjecture (that a decomposition of the vertex/loop skew brace L_ζ suffices for a decomposition of G) is *false*:

Theorem 1.9 (Vertex splitting is necessary but not sufficient, [PROVED]). *There is a connected quiver skew brace on $V \times \text{Coarse}(\{0, 1\})$ with V the Klein four-group and star group $(\mathbb{Z}/2)^3$, and a proper ideal sub-bundle D , such that the vertex skew brace L_ζ decomposes (D_ζ has a sub-skew-brace complement) yet D has no wide sub-quiver-skew-brace complement, so $G \not\cong (G/D) \rtimes D$.*

The correct invariant is a *star complement*: $G \cong (G/D) \rtimes D$ iff there is a multiplicative complement K of D_ζ in L_ζ together with connecting labels (τ_μ) (with $\tau_\zeta = e$) such that $A = \{(\zeta, \mu, k\tau_\mu) : k \in K, \mu \in X\}$ is an additive complement of D_ζ closed under $+_\zeta$ in the whole base star. Closure on one base star forces closure on every star, via the additive isomorphisms $\rho_a = a \triangleright (-)$ carrying D_ζ to D_μ ; so the obstruction is base- ζ localised. For the Klein witness all single-operation shadows split — additive extension of the star, additive complement of D_ζ , multiplicative complement $K = \{0, 3\}$, and the reduced base extension — yet no star complement exists. This is the same both-split-no-common-complement phenomenon as W_4 , now with the extra lesson that testing the loop shadow instead of the base star is precisely the mistake the heap-transport conjecture made. (A brute-force sweep of the Klein family found 504 vertex-splitting-but-not-decomposing configurations and 0 wide complements; the star-transport lemma held in 1008/1008 cases.)

1.5 The internal residue is an orthogonal factor

We now separate $[\Omega]$ from the datum most easily mistaken for part of it: the residue $[\nu]$, which records how the internal action λ relabels D along a section. All of this section is [PROVED].

Lemma 1.10 (Torsor lemma). *As s' ranges over all normalised sections, $(\nu'_h)_{h \neq 0}$ ranges over exactly $T_\nu := \prod_{h \neq 0} \nu_h L$, a torsor under the free transitive action of $L^{H \setminus 0}$. Explicitly $\nu'_h = \nu_h \circ \lambda_{e(h)}|_D$ with $e(h) = \lambda_{s(h)}^{-1}(\theta(h))$.*

Lemma 1.11 (Residue invariance). *The cosets $[\nu_h] = \nu_h L \in \text{Aut}(D, +)/L$ are section-independent, and $\nu_{h_1} \nu_{h_2} = \nu_{h_1 \circ h_2} \cdot \lambda_{\bar{\tau}(h_1, h_2)}|_D$ with $\lambda_{\bar{\tau}}|_D \in L$. When $L \trianglelefteq \text{Aut}(D, +)$ (automatic if $\text{Aut}(D, +)$ is abelian, e.g. D cyclic), $[\nu]: (H, \circ) \rightarrow \text{Aut}(D, +)/L$ is a homomorphism.*

Lemma 1.12 (Obstruction ν -freeness / descent, “fact B”). *With $(D, +), (D, \circ)$ abelian, $[\beta], [\bar{\tau}]$ and the predicate “ $[\Omega] = 0$ ” are constant on the torsor T_ν and refer to no ν . Concretely the section-shifts $(\beta' - \beta) = d\theta$ and $(\bar{\tau}' - \bar{\tau}) = d_\sigma e$ involve no ν .*

Theorem 1.13 (Orthogonal direct factor, [PROVED]). *The section-independent invariants of the extension are $(\mu, \sigma, [\nu], [\Omega])$, and the residue $[\nu] = ([\nu_h])_h \in \prod_h \text{Aut}(D, +)/L$ is orthogonal to $[\Omega]$:*

1. *its section ambiguity is a full torsor T_ν over $L^{H \setminus 0}$ with invariant residue $[\nu]$ — an “ H^1 over L ”;*
2. *$[\beta], [\bar{\tau}]$ and splitting are ν -free and constant on T_ν , and $[\nu]$ is defined without reference to $\beta, \bar{\tau}$ or the ambient defect.*

It is worth being exact about *which* orthogonality holds, because a weaker statement is false and the distinction is the crux.

Proposition 1.14 (Not a gauge-orbit product — the sharp negative, [PROVED]). *The gauge action of the section group on the cochain triple $(\nu, \beta, \bar{\tau})$ is not a direct product of a ν -action and an obstruction-action: $\ker(\Psi_{\text{obs}}) + \ker(\Psi_\nu) \neq \Theta$. For the braces **sb#11**, **sb#15** on $\mathbb{Z}/2 \times \mathbb{Z}/4$ with $D = \{0\} \times \mathbb{Z}/4$ ($L = \text{Aut}(D, +) = \mathbb{Z}/2$), any section that flips ν_1 forces $d\theta(1, 1) = (0, 2) \neq 0$.*

That is: at the level of *orbits* (raw cochains), moving the residue is coupled to a nonzero change in the obstruction cochain. But that change is a coboundary — invisible to cohomology. Orthogonality holds for *classes*, and fails for orbits. The independence theorem is a statement about classes; the negative proposition is the fence that stops us over-claiming it.

Non-trivial kernels do not disturb $[\Omega]$. Dropping (SH) — allowing D to be a non-trivial brace — changes the residue’s behaviour but not the obstruction:

Theorem 1.15 (Kernel non-triviality is orthogonal, [PROVED]). *For an ideal D with $(D, +)$ abelian, ν is section-independent for all sections iff $\lambda_e|_D = \text{id}$ for all $e \in D$ iff D is a trivial brace. When D is a non-trivial brace, the sole effect is to make ν section-dependent; the descent maps $\varphi_+, \varphi_\circ$ and the ambient common-complement defect are unchanged. Explicitly, for a lift k of an order-2 quotient, $k \circ k = (k + k) + (\lambda_k - \text{id})(k)$, and $G \cong H \bowtie D$ iff there is $k \in A := \{k : k + k = 0\}$ with $(\lambda_k - \text{id})(k) = 0$; the defect $(\lambda_k - \text{id})(k)$ uses λ_k for a lift $k \notin D$, an ambient feature into which ν never enters.*

The witness makes the orthogonality vivid: on $\mathbb{Z}/2 \times \mathbb{Z}/4$ with $(D, +) \cong \mathbb{Z}/4$, the non-trivial-brace kernels **sb#11**, **sb#15** (with $(D, \circ) \cong \mathbb{Z}_2^2$) and the trivial-brace **sb#1** share the same $A = \{(1, 0), (1, 2)\}$, the same defect $(0, 2) \neq 0$, hence the same $[\beta] = [\bar{\tau}] = 0$ and $[\Omega] \neq 0$; they differ *only* in the number of realisable ν_1 (1 versus 2). Kernel non-triviality is carried entirely by ν , and ν never touches $[\Omega]$.

1.6 The product theorem

We can now upgrade orthogonality of a single pair of invariants to a global product decomposition of the moduli of extensions. Fix the additive group $(G, +)$, the subgroup D (hence μ and the additive class $[\beta]$), the D -brace $(L, \lambda|_D)$, the H -brace $(H, +, \circ_H)$, and the circle action σ . Among all skew-brace structures $\circ$ on $(G, +)$ realising this data with D an ideal, taken up to equivalence of extensions, let R be the set of *realised* residues $[\nu]$.

Theorem 1.16 (Residue-independent obstruction group; product, [PROVED]).

1. *For every $r \in R$, the equivalence classes of extensions with residue r form a torsor under an abelian group Ω^0 that is independent of r .*
2. *Consequently $([\nu], [\Omega]) : \text{Ext} \rightarrow R \times \Omega^0$ is a bijection: every realised residue co-occurs with every value of $[\Omega]$.*

Proof (four steps). Step 1 (the residue side is T -free). By (coupling-1), the admissible residues are cut out using only ν, μ, λ^H — never T or β . This is block-triangularity: the $D \rightarrow D$ block of λ , carrying the residue, satisfies a closed system not involving the off-diagonal obstruction block.

Step 2 (for fixed ν , the valid T form an $\emptyset$ -or-torsor under a ν -free group Z^0). Read (c) as an equation for T with $\beta, \mu, \nu, \lambda^H$ fixed. It is *affine* in T , with homogeneous part

$$U(h, k +_H l) = U(h, k) + \mu_{(h \circ k) - h} U(h, l). \quad (c^0)$$

The crux is that $(h \circ k) - h = \lambda_h^H(k)$ in H , so the twist $\mu_{(h \circ k) - h} = \mu_{\lambda_h^H(k)}$ depends only on μ and λ^H — *not on ν* . Together with the homogeneous circle-cocycle condition (b^0) (also ν -free), set $Z^0 := \{U : H \times H \rightarrow D \mid (b^0) \wedge (c^0)\}$, manifestly independent of ν and β . If T, T' both solve $(b) \wedge (c)$ then $U := T - T' \in Z^0$ (the $\mu_h \nu_h \beta$ term and all β -terms cancel on subtraction), and if T solves $(b) \wedge (c)$ and $U \in Z^0$ then $T + U$ does. So the solution set $\mathcal{T}(\nu)$ is empty or a torsor under the ν -free group Z^0 .

Step 3 (coboundaries are ν -free; the obstruction group). A change of section shifts T by a coboundary from a subgroup $B^0 \subseteq Z^0$, shifts ν within its L -coset (residue preserved, Lemma 1.11) and shifts β by an additive coboundary. The T -coboundaries B^0 are ν -free (Lemma 1.12). Put $\Omega^0 := Z^0/B^0$. For each realised residue r , the equivalence classes of extensions with residue r are a torsor under Ω^0 , and since Z^0, B^0 are ν -free, Ω^0 is residue-independent. This is (1).

Step 4 (the product). By (1), for each realised r the class $[\Omega]$ ranges over the entire Ω^0 as the extension ranges over the full torsor $\text{Ext}(r)$. Fixing a base-point identification $\text{Ext}(r) \cong \Omega^0$, every value of Ω^0 is attained for every realised r , and because each $\text{Ext}(r)$ is a *full* torsor the conclusion is base-point independent. Thus $([\nu], [\Omega])$ is a bijection onto $R \times \Omega^0$. $\square$

Nothing in the proof uses $[\beta] = 0$: the theorem holds for every additive class. The reader should note where each hypothesis is spent. That $(D, +), (D, \circ)$ are abelian gives the factor-set calculus (a)–(c); fixing the H -brace λ^H makes the twist in (c^0) ν -free; fixing μ, σ makes (b^0) and B^0 ν -free. This is the “internal-replacement” statement in its cleanest form: *changing the internal relabelling action $[\nu]$ does not change which decomposition obstructions $[\Omega]$ are available*. They are independent coordinates of the moduli.

Corollary 1.17 (Split $\perp$ residue, [PROVED]). *The split value $[\Omega] = 0$ is realisable iff $[\beta] = 0$ (additive split), uniformly in the residue: a sub-skew-brace complement is a fortiori an additive complement, so $[\Omega] = 0 \Rightarrow [\beta] = 0$; conversely $[\beta] = 0$ makes every admissible residue realisable by a bicrossed product, which is split. Hence when $[\beta] = 0$, $\text{Ext} \cong R \times \Omega^0$ with split and non-split each a residue-independent subset; when $[\beta] \neq 0$ no extension is split, and the realisable $[\Omega]$ -values are the non-zero classes, attained uniformly across all realised residues.*

1.7 A genuine $|G| = 16$ witness, and the one open step

The independence is easy to believe — and easy to dismiss as a small-example artifact. When $|D| = 4$ one has $\text{Aut}(\mathbb{Z}/4, +) = \mathbb{Z}/2$, so $L \neq 1$ forces $L = \text{Aut}(D, +)$; the residue then has no room to be a *proper* coset, and the coupling never bites. A convincing witness must have $L \neq 1$ and $L \subsetneq \text{Aut}(D, +)$, which needs $|D| \geq 8$, hence $|G| \geq 16$. This corner is where the phenomenon is real, and it is [COMPUTED].

Example 1.18 (The $|G| = 16$ corner, [COMPUTED]). Take $(G, +) = \mathbb{Z}/2 \times \mathbb{Z}/8$, $D = \{(0, x)\} \cong \mathbb{Z}/8$, $H = \mathbb{Z}/2$, so $\text{Aut}(D, +) = (\mathbb{Z}/8)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. Use the unit-scaling family $\varphi_{u,t}(j, y) = (j, u \cdot y + j \cdot t)$ and the homomorphism $U(i, x) = 5^x \bmod 2 \cdot 3^i \pmod{8}$. Two structures differ only in a part invisible on D (so they share L and ν):

$$G_0 : \lambda_{(i,x)}^0 = \varphi_{U(i,x),0} \quad G_1 : \lambda_{(i,x)}^1 = \varphi_{U(i,x),4i},$$

where $T(i, x) = 4i$ is a homomorphism $(G, \circ) \rightarrow \{0, 4\}$. Then $L = \{1, 5\}$ is a *proper, non-trivial* subgroup, $1 \subsetneq L \subsetneq \text{Aut}(D, +)$; the base lift $s(1) = (1, 0)$ gives $\nu_1 = (\cdot 3) \notin L$ and residue $[\nu_1] = \{3, 7\} \neq L$. Both structures carry the *same* $(L, [\nu]) = (\{1, 5\}, \{3, 7\})$, yet $[\Omega] = 0$ for G_0 (the complement $\{(0, 0), (1, 0)\}$ is closed under both operations) and $[\Omega] \neq 0$ for G_1 (every order-2

candidate fails $\circ$ -closure: $(1, x) \circ (1, x) = (0, 4)$). So independence persists at genuine moving ν with non-trivial residue — it is not a $|D| = 4$ size artifact.

The moving- ν cocycle equations of Example 1.18 are exactly (a),(b),(c) of §1.2, machine-checked on the witness; the mechanism verdict is the one already used in the product theorem. At the *cochain* level there is a genuine coupling — moving ν_h inside its L -coset forces a compensating change in T . At the *invariant* level there is independence, because $a \circ b = a + \lambda_a(b)$ means that once $(G, +)$ (hence μ, β) and λ (hence $[\nu]$ and the L -part) are fixed, $\circ$ and therefore T are *determined*; equation (c) merely *defines* T from β up to a circle-coboundary.

regime	additive group	$[\beta]$	result
product enum	$\mathbb{Z}/2 \times \mathbb{Z}/8$	0	every σ : full product of $([\nu], [\Omega])$; counts balanced
equiv-classes	$\mathbb{Z}/2 \times \mathbb{Z}/8$	0	each admissible residue: exactly $2 L $ classes ($ L $ split, $ L $ non-split)
equiv-classes	$\mathbb{Z}/16$	$\neq 0$ (ord 2)	each residue: $2 L $ classes, all non-split (uniform)
equiv-classes	$\mathbb{Z}/32$	$\neq 0$ (ord 4)	residues 1, 3, 5, 7 all realised, each with exactly 4 classes
engine-verify	$\mathbb{Z}/32$	$\neq 0$	cross-equation (c) holds for all 28 ideals ($H = \mathbb{Z}/4$)

Table 1.1: Exact enumeration corroborating Theorem 1.16. The constant equivalence-class count per residue within each (σ, L) block is the prediction $|\text{Ext}(r)| = |\Omega^0|$. The $[\beta] \neq 0$, order-4 case ($\mathbb{Z}/32$) realises all four admissible residues, refuting any fear that $(\chi - 1)_*[\beta] \neq 0$ could block existence.

The one open step, stated precisely. Theorem 1.16 (the product on *realised* residues) is unconditional. What is [COMPUTED] but not [PROVED] is a single surjectivity statement:

Surjectivity of the residue map for $[\beta] \neq 0$: every coupling-admissible residue is actually realised (i.e. $R = R_{\text{adm}}$) when the additive extension is non-split.

By Steps 2–3 this is equivalent to $\mathcal{T}(\nu) \neq \emptyset$ for every admissible ν , i.e. to the assertion that the RHS of (c) is always a T -coboundary (equivalently: associativity of $\circ$ forces (c) to be solvable for T , uniformly in the admissible residue). It is verified on $\mathbb{Z}/16$ and $\mathbb{Z}/32$ (all admissible residues occur) and argued structurally on the witness, but not closed in general. Three things should be emphasised, in the interest of not over-claiming:

- the open residual concerns the *residue map alone* — it is not about the coupling term, and not about the product structure, both of which are established;
- it does not affect Theorem 1.16, which is a statement about the realised set R ;
- the exact (σ, L) -block grouping is clean only for $H = \mathbb{Z}/2$ (where σ is fully sampled at the single non-zero h); for $H = \mathbb{Z}/4$ the equivalence-class table is corroborated under a coarser σ -grouping, while the (c)-formula check there is exact.

Closing this residual is the business of a proof session, not this chapter; it is flagged in the honest ledger below.

1.8 Honest ledger, and Ferri’s Remark 3.20

Statement	Grade
$[\Omega] = [(\beta, \bar{\tau})] \in H_{\text{sb}}^2(H, D)$ is the Rathee–Yadav class; φ_+ descends; $[\Omega] = 0 \iff$ bicrossed (trivial-kernel SH)	[PEER-REVIEWED]
Product map $\Phi = (\varphi_0, \varphi_+)$ non-injective (W_4)	[PEER-REVIEWED]
$[\Omega]$ is a bilinear class, strictly finer than both group Schreier classes; base- ζ localised; heap-transport refuted	[PROVED]
$[\nu]$ is an orthogonal direct factor; gauge-orbit product is <i>false</i> , orthogonality holds at the invariant level	[PROVED]
Non-trivial-brace kernels do not disturb $[\Omega]$ (carried by ν)	[PROVED]
Product theorem $\text{Ext} \cong R \times \Omega^0$, Ω^0 residue-independent (on realised residues; unconditional)	[PROVED]
$ G = 16$ genuine- L corner; moving- ν (c) machine-checked	[COMPUTED]
Surjectivity of the residue map for $[\beta] \neq 0$ (every admissible residue realised)	[COMPUTED]
φ_+ descent and W_4 non-injectivity decide certificates	[LEAN-VERIFIED]

Two honesty notes are load-bearing and belong on the record.

- *No socle-necessity claim.* An earlier draft of the identification asserted that separation “requires $D \not\subseteq \text{Soc}(G)$.” On referee review (Rick) this was found to be false — a trivial-kernel brace with $D = \text{Soc}(G)$ exactly nonetheless separates — and the claim is *deliberately omitted* here. It is not needed for any result in this chapter.
- *The general kernel is native.* The identification of $[\Omega]$ with a published cohomology (Rathee–Yadav) is proved only for the trivial-kernel standing hypothesis. Housing the general non-trivial-kernel obstruction in a skew-brace H^2 where ν is *varying* cocycle data is native to this programme and not yet reduced to any published theory; we do not claim otherwise.

Relation to Ferri’s Remark 3.20. Ferri [1] asks, in the language of quiver skew braces, for a group-theoretic reformulation of when an ideal is *prunable* — when the quiver skew brace decomposes off that ideal. Theorem 1.7 supplies exactly this: prunability of D is the vanishing of the single skew-brace class $[\Omega]$, equivalently the existence of a sub-skew-brace (star) complement; and Theorem 1.2 explains *why* no purely group-theoretic (single-operation) criterion can suffice — the obstruction is bilinear, and lives in the coupling that no forgetful functor to a group H^2 can retain. The base-localisation of Theorem 1.9 is the groupoid refinement: prunability is decided on one star and transported.

What this feeds. For the grant, the chapter closes the skew-brace / Yang–Baxter instance of the decomposition programme with a clean separation of concerns: an *existence* axis ($[\omega]$, Chapter ??) and a strictly finer *decomposition* axis ($[\Omega]$, this chapter), the latter a bilinear invariant that is provably independent of the internal relabelling residue. The independence is the mathematical content of “internal replacement”: one may re-index the internal action of a composite system without changing which structural decompositions are available to it. The single open surjectivity

step is isolated, small, and does not touch the product structure — it is the natural target of the next proof session.

Bibliography

- [1] D. Ferri, *On quiver skew braces, their ideals and products*, arXiv:2605.11903, 2026.
- [2] N. Rathee, M. K. Yadav, *Skew brace extensions, second cohomology and complements*, arXiv:2601.12371, 2026.